

No. of Printed Pages : 4

MCS-224

**MASTER OF COMPUTER
APPLICATIONS (MCA-NEW)**

Term-End Examination

June, 2022

**MCS-224 : ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

***Note :** Question No. 1 is compulsory. Attempt any
three questions from the remaining
questions.*

1. (a) Compare Narrow AI, General AI and Super AI. Give suitable example for each. 6
- (b) Briefly discuss the formulation of state space search, with a suitable example. 5

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- (c) Write DFS (Depth-First Search) algorithm. Give **one** advantage and **one** disadvantage of DFS. 5
- (d) Obtain Disjunctive Normal Form for the well form formula given below : 5
- $$\sim (A \rightarrow (\sim B \wedge C))$$
- (e) Briefly discuss the relevance of resolution and unification mechanism in artificial intelligence. 4
- (f) Compare supervised learning and unsupervised learning. List the algorithms for each type of learning. 5
- (g) Write Bayes' theorem and elaborate the meaning of each component in it. 5
- (h) What is Dimensionality Reduction ? Briefly discuss the ways to achieve it. 5
2. (a) What do you understand by the term "Agents" in Artificial Intelligence ? List the properties, supposed to be possessed by agents. Also, compare the SR (simple reflex) agents with model based reflex agents. 10

- (b) Draw the block diagram of machine learning cycle and briefly discuss the role of each of its components. 10
3. (a) What is Adversarial Search ? How it is different from the normal search ? Briefly discuss the types of adversarial search. 10
- (b) Differentiate between the following, with an example for each : 10
- (i) Machine learning and Data mining
 - (ii) Multi-class classification and Multi-label classification
4. (a) What is Best First Search ? How is it different from greedy best first search ? Give time complexity and space complexity of best first search. Also, give advantage and disadvantage of best first search. 10
- (b) How regression differs from classification ? What is the similarity between the two ? Briefly discuss the following type of regression : 10
- (i) Linear regression

- (ii) Multiple linear regression
- (iii) Logistic regression
- (iv) Polynomial regression

5. Write short notes on any **five** of the following :

5×4=20

- (a) Rule based systems and its types
- (b) Frames
- (c) Dempster Scheffer theory
- (d) Closed world assumption
- (e) Recursive neural networks
- (f) Generative adversarial networks
- (g) Support vector machines
- (h) Confusion matrix
- (i) Linear discriminant analysis
- (j) Pincer search

**MASTER OF COMPUTER
APPLICATIONS (MCA) (NEW)**

Term-End Examination

December, 2022

**MCS-224 : ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

***Note : Question No. 1 is compulsory. Attempt any
three questions from the rest.***

1. (a) Compare Artificial Intelligence (AI), Machine Learning and Deep Learning. 6
- (b) Briefly discuss the Adversarial search. Name the techniques used for adversarial search. 5
- (c) Write algorithm for BFS (Breadth-First Search). Write the time complexity and space complexity of BFS. 5

- (d) Obtain Conjunctive Normal Form (CNF) for the formula : 5

$$D \rightarrow (A \rightarrow (B \wedge C))$$

- (e) What is Skolemization ? Skolemize the expression : 4

$$(\exists_{X_1})(\exists_{X_2})(\forall_{Y_1})(\forall_{Y_2})(\exists_{X_3})(\forall_{Y_3})$$

$$P(X_1, X_2, X_3, Y_1, Y_2, Y_3)$$

- (f) What is Reinforcement Learning ? Classify the various reinforcement learning algorithms. 5
- (g) What is Logistic Regression ? Briefly discuss the various types of logistic regressions. 5
- (h) Differentiate between linear regression and polynomial regression techniques. 5
2. (a) In context of Intelligent Agents, what are task environments ? Explain the standard set of measures for specifying a task environment under the heading PEAS. 10
- (b) Briefly discuss the following (give suitable example for each) : 10
- (i) Rote learning
 - (ii) Supervised learning
 - (iii) Unsupervised learning
 - (iv) Delayed-Reinforcement learning

3. (a) Briefly discuss the Min-Max Search Strategy. What are the properties of Minimax Algorithm ? Also give advantages and disadvantages of Minimax search. 10
- (b) Differentiate between the following, with an example for each : 10
- (i) Classification techniques and Regression techniques
- (ii) Lazy learner algorithms and Eager learner algorithms
4. (a) What is Iterative Deepening Depth First Search (IDDFS) ? How is it different from Depth First Search ? Give time and space complexities of IDDFS. Also give advantages and disadvantages of IDDFS. 10
- (b) Discuss support vector regression. Draw suitable diagram in support of your discussion. Also give *two* applications of support vector regression. 10
5. Write short notes on any *five* of the following : 5×4=20
- (a) Forward Chaining
- (b) Semantic Nets

- (c) Bayes' Networks
- (d) Rough Set Theory
- (e) Recurrent Neural Networks
- (f) Restricted Boltzmann Machines
- (g) Ensemble Methods
- (h) K-Nearest Neighbour
- (i) Principal Component Analysis
- (j) Association Rules

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**MASTER OF COMPUTER
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Term-End Examination

June, 2023

**MCS-224 : ARTIFICIAL INTELLIGENCE AND
MACHINE LEARNING**

Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : *Question No. 1 is compulsory. Attempt any
three questions from the remaining
questions.*

1. (a) Compare descriptive, predictive and prescriptive analytics in machine learning.

6

- (b) What is Min-Max Search Strategy ? Write MINIMAX algorithm.

5

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(c) Differentiate between informed search and uninformed search. Name *one* algorithm for each. 4

(d) Describe the Modus Ponens and Modus Tollens as propositional rule of inference. 5

(e) What is Prenex Normal Form (PNF) ? Transform the following formula into PNF : 5

$$(\forall_x)(Q(x) \rightarrow (\exists_x)R(x, y))$$

(f) What is Ensemble Learning ? Briefly discuss any *one* of the ensemble learning method. 5

(g) Draw confusion matrix and write formula for accuracy, precision, sensitivity and specificity. 5

(h) What is a Neural Network ? How biological neuron relates to Artificial Neuron ? Illustrate with suitable diagram and a table to map the components of Biological Neuron with Artificial Neuron. 5

2. (a) Explain Turing test, with the help of a block diagram. Also, discuss Chinese room test as criticism to Turing test. 10
- (b) Briefly discuss the following, with suitable example for each : 10
- (i) Rule-based machine learning
 - (ii) Bayesian Algorithms
 - (iii) Decision trees
 - (iv) Dimensionality reduction
3. (a) What do you understand by State Space Search ? Explain the state space representation of Water-Jug Problem (WJP), given below :
- “Given two jugs of 5-gallon and 3-gallon, both of which do not have measuring indicators on them. The jugs can be filled with water with the help of any pump, any number of times.”
- The question is “how can you get 4 gallons of water in a 5-gallon jug ?” 10

- (b) What is Binary Classification ? Can binary classification algorithms be altered to work for problems with more than two classes ? Justify. Also, discuss one-versus the rest and one-versus-one approach. 10
4. (a) Differentiate between the following : 10
- (i) A* and AO* algorithm
 - (ii) Depth first search and Breadth first search
- (b) What is linear regression ? How linear regression is performed using least square method ? Find the regression line for the data points (x, y) tabulated below : 10

x	y
1	3
2	4
3	2
4	4
5	5

Also, discuss the terms 'mean squared error' and 'mean absolute errors'.

5. Write short notes on any *five* of the following :

5×4=20

- (a) Backward Chaining
- (b) Scripts
- (c) Non-Monotonic Reasoning Systems
- (d) Convolutional Neural Networks
- (e) Auto Encoders
- (f) Transformers
- (g) Logistic Regression
- (h) Naive Bayes Algorithm
- (i) Feature Selection
- (j) K-Means Algorithm

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Note : *Question No. 1 is compulsory. Attempt any
three questions from the rest.*

1. (a) Differentiate between clustering and classification technique. List any *two* algorithms for each. 6

- (b) Explain Turing test with suitable example.

5

- (c) What is skolemization ? What is the utility of skolemization ? Skolemize the following expression : 6

$$\exists x_1 \exists x_2 \forall y_1 \forall y_2 \exists x_3 \forall y_3 P(x_1, x_2, x_3, y_1, y_2, y_3)$$

- (d) Explain backward chaining system with a suitable example. 6

- (e) Differentiate between Lazy learners and Eager learners in classification problem. Also, list the algorithms used for each type of learners, respectively. 6

- (f) Explain the term 'Dimensionality Reduction'. Name the general techniques used to perform it. Also give merits and limitation of dimensionality reduction. 6

- (g) Explain the working of partition based clustering. Mention any *two* methods used for partition based clustering. 5
2. (a) Compare artificial intelligence, machine learning and deep learning. 6
- (b) What do you understand by state space in AI ? What is its utility ? Write production rules for state space representation of water jug problem. 7
- (c) Write and explain Breadth First Search (BFS) algorithm. Discuss its space and time complexity. Also, give advantage and disadvantage of BFS algorithm. 7
3. (a) Differentiate between predicate and propositional logic. If $P(x) \rightarrow$ “ x is a rational number” and $Q(x) \rightarrow$ “ x is a real

number” then symbolize the following sentences :

5

(i) Every rational number is a real number

(ii) Some real numbers are rational

(iii) Not every real number is a rational number

(b) Explain the concept of resolution and unification in AI, with suitable example for each.

7

(c) Explain rule based systems in AI. Give advantages and disadvantages of rule based systems. Also, give the *two* important sources of uncertainty in rule based systems.

8

4. (a) Explain reinforcement learning with the help of a block diagram. Explain the role of each component of block diagram. 6
- (b) Compare classification and regression techniques of supervised learning. Explain the various metrics used for evaluating the classification model. 7
- (c) Differentiate between the following (give example for each) : 7
- (i) Logistic regression and Linear regression
- (ii) K-NN algorithm and K-means algorithm
5. Explain any ***four*** of the following with suitable example for each : $4 \times 5 = 20$
- (a) Linear Discriminant Analysis

- (b) FP tree growth
- (c) Density based clustering
- (d) Restricted Boltzmann Machines
- (e) Convolutional Neural Networks

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MCS-224

**MASTER OF COMPUTER
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June, 2024

**MCS-224 : ARTIFICIAL INTELLIGENCE AND
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Time : 3 Hours

Maximum Marks : 100

Weightage : 70%

Note : *Question No. 1 is compulsory. Attempt any
three questions from the rest.*

1. (a) Compare Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI). 6

P. T. O.

- (b) Explain Chinese room test as Criticism to Turing test, with suitable example. 5
- (c) Write steps to transform FOPL (First Order Predicate Logic) to PNF (Prenex Normal Form). Apply the steps to transform $\forall_x(Q(x) \rightarrow \exists_x R(x, y))$ to PNF. 6
- (d) Explain forward chaining systems with suitable example. 6
- (e) Draw block diagram for the machine learning cycle. Also list the steps involved in machine learning cycle. 6
- (f) Explain the working of FP growth algorithm. Give advantages of FP growth over Apriori Algorithm. 6
- (g) Compare clustering and classification. Give suitable example for each. Also, list the algorithms for each. 5

2. (a) Compare descriptive, predictive and prescriptive analytics, performed under machine learning. 6
- (b) Explain Min-Max search algorithm with suitable example. Give properties of Min-Max search algorithm. Also, write its advantages and disadvantages. 7
- (c) Write and explain Depth First Search (DFS) algorithm. Give time and space complexity of DFS algorithm. Also give its advantage and disadvantage. 7
3. (a) Given $C(x)$ = "X is a used car dealer", and $H(x)$ = "X is honest" then translate the following into English sentences : 5
- (i) $\exists_x C(x)$
- (ii) $\exists_x H(x)$
- (iii) $\forall_x C(x) \rightarrow \sim H(x)$
- (iv) $\exists_x (C(x) \wedge H(x))$
- (v) $\exists_x (H(x) \rightarrow C(x))$

- (b) What do you understand by the term 'Resolution' in AI ? Discuss the utility of Resolution mechanism in AI. Apply it to conclude 'Raman is mortal' from the knowledge given below : 7

(i) Every man is mortal

(ii) Raman is a man

- (c) Write short notes on any *two* of the following, with suitable example for each : 8

(i) Frames

(ii) Scripts

(iii) Semantic nets

4. (a) Compare model-free reinforcement learning with mode-based reinforcement learning. Also, discuss the sub-classes of model-free reinforcement learning. 6

- (b) Write and explain Bayes' theorem. Also, write and explain Naive Bayes' algorithm with suitable example. 7
- (c) Write short notes on any *two* of the following (give example for each) : 7
- (i) Support Vector Machines
 - (ii) Support Vector Regression
 - (iii) Polynomial Regression
5. Explain any **four** the following with suitable example for each : 4×5=20
- (a) Principal Component Analysis
 - (b) Apriori Algorithm
 - (c) Hierarchical Clustering
 - (d) Generative Adversarial Networks
 - (e) Auto Encoders